

Praveen Kumar

Technical Architect | Leeds, United Kingdom

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PROFESSIONAL SUMMARY

Technical Architect who builds and ships AI-native products end-to-end architecture, backend, database, and LLM/agent layers as an Azure Solutions Architect & Azure AI Engineer Associate delivering enterprise-scale cloud and AI platforms across the UK and India. Designs and deploys production LLM workflows and multi-agent systems (RAG pipelines, Foundry Agent Service, LangGraph, Microsoft Agent Framework, MCP server integrations) alongside the underlying cloud infrastructure (Azure PaaS, Infrastructure as Code via Terraform, observability with Azure Monitor/Application Insights), several shared as [open-source](#) reference projects. Delivered complex Azure and AI solutions for FTSE 100 clients including [ASDA](#), [BP](#), [AON](#), [DLL](#), [CAPITA](#), [UnitedHealth Group](#), and [Airbus](#), driving faster time-to-production for cloud and AI platforms, underpinned by delivery of high-quality web and API products at scale (27M+ monthly users) using Azure PaaS, Infrastructure as Code (Terraform), and observability tooling (Azure Monitor/Application Insights). Passionate about clean, testable code, TDD, security best practices, and cross-functional collaboration in Agile teams to deliver user-focused solutions that drive business value.

CORE SKILLS

- **Agentic AI & Foundry:** Microsoft Foundry, Foundry Agent Service, Agentic Workflows, Multi-Agent Pipelines, LangChain, LangGraph, LangSmith, AutoGen, CrewAI, Semantic Kernel, MCP, Microsoft Agent Framework (MAF).
- **Generative AI & LLMs:** RAG, Prompt Engineering, Fine-Tuning, Vector/Semantic Search, Embeddings, LLM Evaluation, Red Teaming, Guardrails, Azure AI Content Safety, Agentic AI, Model Context Protocol (MCP), Retrieval-Augmented Generation Pipelines, LLM Orchestration, Chunking Strategies, Hybrid Search, Model Benchmarking, Hallucination Mitigation, Token Optimization, Multimodal AI (Vision + Language), Function/Tool Calling.
- **Azure AI Services:** AI Search, Document Intelligence, Vision, Speech, Language, Translator, Cognitive Services.
- **Azure Cloud & Architecture:** Azure Functions, API Management, App Services, Cosmos DB, Event Hub, Service Bus, ACR, ACI, Azure Firewall, Application Gateway, VNET/NSG, Key Vault, Storage, Microsoft Entra ID, Azure Kubernetes Service (AKS), Azure SQL Database, Azure Monitor, Application Insights, Log Analytics, Logic Apps, Event Grid, Azure Load Balancer, Traffic Manager, Managed Identities, Azure Policy, Cost Management, Well-Architected Framework.
- **Engineering & DevOps:** Python, C#, .NET Core, T-SQL, JavaScript, Azure DevOps, Terraform, Jenkins, CI/CD, Containerization, Observability.
- **Architecture & Delivery:** Microservices, Serverless, Event-Driven, Enterprise Architecture, Agile, Pre-Sales, Stakeholder Management.
- **Web Application Development:** APIs, JavaScript, HTML, CSS, jQuery.

PROFESSIONAL EXPERIENCE

Technical Architect — [TCS](#), United Kingdom

Apr 2021 – Present

- Architected a scalable Azure PaaS platform for [ASDA](#), supporting the separation of 2,500+ systems from Walmart and improving e-commerce throughput by 30% through event-driven microservices and Azure API Management (APIM), exposing high-traffic applications for public and private access with 99% uptime and a 35% efficiency gain, driving a 28% boost in user engagement and a 42% reduction in page load times.
- Built demand-forecasting and inventory/stock-prediction models for ASDA's retail operations, feeding predictive outputs into downstream Azure-hosted planning workflows to support stock availability decisions.
- Designed and delivered RAG-based conversational AI systems using Microsoft Foundry and Azure AI Search and built multi-agent AI pipelines with LangGraph and Foundry Agent Service, including supervisor routing, retry logic, and LangSmith observability.
- Drove Microsoft Purview-integrated metadata ingestion pipelines for [BP](#), enabling automated data lineage and governance across 12 data sources and 4,000+ datasets, improving data discoverability and reducing manual governance effort across distributed data estates.
- Guided architecture workshops, technical reviews, and Agile delivery teams across multiple concurrent client engagements, including [AON](#), mentoring teams on cloud automation practices, reducing cloud spend by 12%, and leading incident response efforts using automated runbooks and root cause analysis (RCA).

Technical Lead — [Wipro Ltd.](#), Pune, India

Jul 2020 – Apr 2021

- Directed cloud modernization for [UnitedHealth Group](#), migrating 4 legacy applications to Azure PaaS using microservices and Azure Functions, reducing infrastructure costs by an estimated 36% and improving API performance by 37% (measured via Application Insights).
- Designed secure APIs using .NET Core and Azure API Management with OAuth 2.0 and RBAC across 2 applications. Automated CI/CD pipelines using Azure DevOps and Terraform, shortening deployment cycle time and improving release reliability.

Consultant — [Atos India Pvt. Ltd.](#), India

Aug 2013 – Mar 2020

- Spearheaded digital transformation programmes for [WSP](#), [DLL](#), and [Airbus](#) using Azure PaaS and .NET Core.
- Built multilingual customer engagement platforms on Sitecore CMS, supporting 10+ languages across client markets.
- Managed Agile teams of 8–12 engineers across design, development, and testing.
- Contributed to pre-sales activities and solution proposals, supporting business development efforts.
- Orchestrated client technical presentations, communicating solution architecture and value to stakeholders.

Senior Software Engineer — [Sopra Group](#), India

Jun 2011 – Jul 2013

- Implemented Azure-based digital platforms for [CAPITA](#) and [Airbus](#), including portals, dashboards, and enterprise integrations, using services such as Azure App Service and Azure Functions.
- Contributed to solution architecture and development, building API-based integrations to connect enterprise systems and meet client requirements.
- Managed development and testing workflows using Azure DevOps within Agile teams, ensuring quality and timely delivery of each release.

Software Engineer — OnDemand Inc. / ODO Software Solutions Pvt. Ltd., India

May 2008 – Jun 2011

- Contributed to the design of ASP.NET applications on project mediafiche, collaborating with Agile teams to define technical solutions.
- Developed and implemented application features, ensuring alignment with project requirements and coding standards.
- Conducted testing (unit/integration) to ensure application quality, reliability, and performance prior to release.

EDUCATION

Master of Computer Applications (MCA) — [United Institute of Management](#), 2008

Bachelor's Degree — [Sachidanand Sinha College](#), 2003

CERTIFICATIONS

- [Azure Solutions Architect Expert \(AZ-305\)](#)
- [Azure Administrator Associate \(AZ-104\)](#)
- [Azure AI Engineer Associate \(AI-102\)](#)

KEY ACHIEVEMENTS

- Delivered 18+ enterprise-grade applications (web, API, batch, console) using Azure PaaS, .NET and Python supporting high-traffic environments (27M+ monthly users), achieving a 99.5% deployment success rate across 160+ Azure resources via Infrastructure as Code and automated CI/CD.
- Reduced critical vulnerabilities by 75% through integrated security scanning (Snyk, Trivy) across hundreds of repositories.
- Created end-to-end infrastructure for 8+ applications covering core Azure services (Microsoft Entra ID, App Services, Azure SQL, APIM, VNet, NSGs, Application Gateway, Key Vault, monitoring, Storage, Service Bus, Event Hub).
- Reduced cloud operational spend by 12% through automation of routine cloud management tasks and mentorship of engineering teams on cost-optimization best practices.
- Improved incident response times by 27% by introducing automated runbooks and standardized root cause analysis (RCA) processes across client engagements.
- Recognized with a TCS Top Performer Award for engineering innovation and technical mentorship across concurrent client programmes.
- Improved RAG response accuracy by 17% through refined chunking, hybrid search, and retrieval evaluation frameworks built on Azure AI Search. Reduced LLM inference costs by 19% via prompt optimization and caching strategies.